

Compilers

- ❑ A compiler is a program that reads a program written in one language, called the source language, and translates it into an equivalent program in another language, the target language.
- ❑ A compiler converts a high level language to machine level language.
- ❑ There are two parts in compilation:
 1. **Analysis:** This part breaks up the source program into constituent pieces and creates an intermediate representation of the source program.
 2. **Synthesis:** This part constructs the desired target program from the intermediate representation.

Analysis

- ❑ In compilation, analysis consists of three phases:
 1. **Linear (lexical) analysis:** In this phase, the stream of characters making up the source program is read from left to right and grouped into tokens that are sequences of characters having a collective meaning.
 2. **Hierarchical (syntax) analysis:** In this phase, characters or tokens are grouped hierarchically into nested collections with collective meaning.
 3. **Semantic analysis:** In this phase, certain checks are performed to ensure that the components of a program fit together meaningfully.

P-Programs

- ❑ It transforms one high level language into another.
- ❑ The task of collecting the source program is sometimes entrusted to a distinct program called a **preprocessor**.

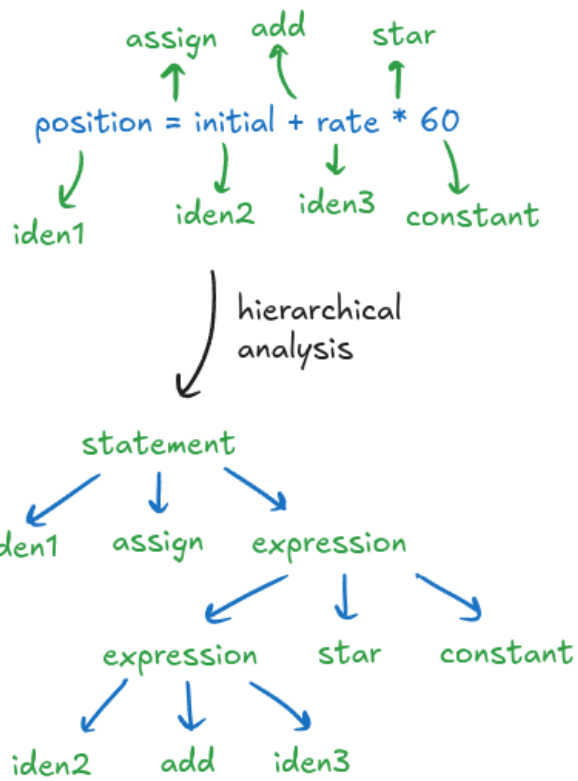


Figure 1: Hierarchical analysis

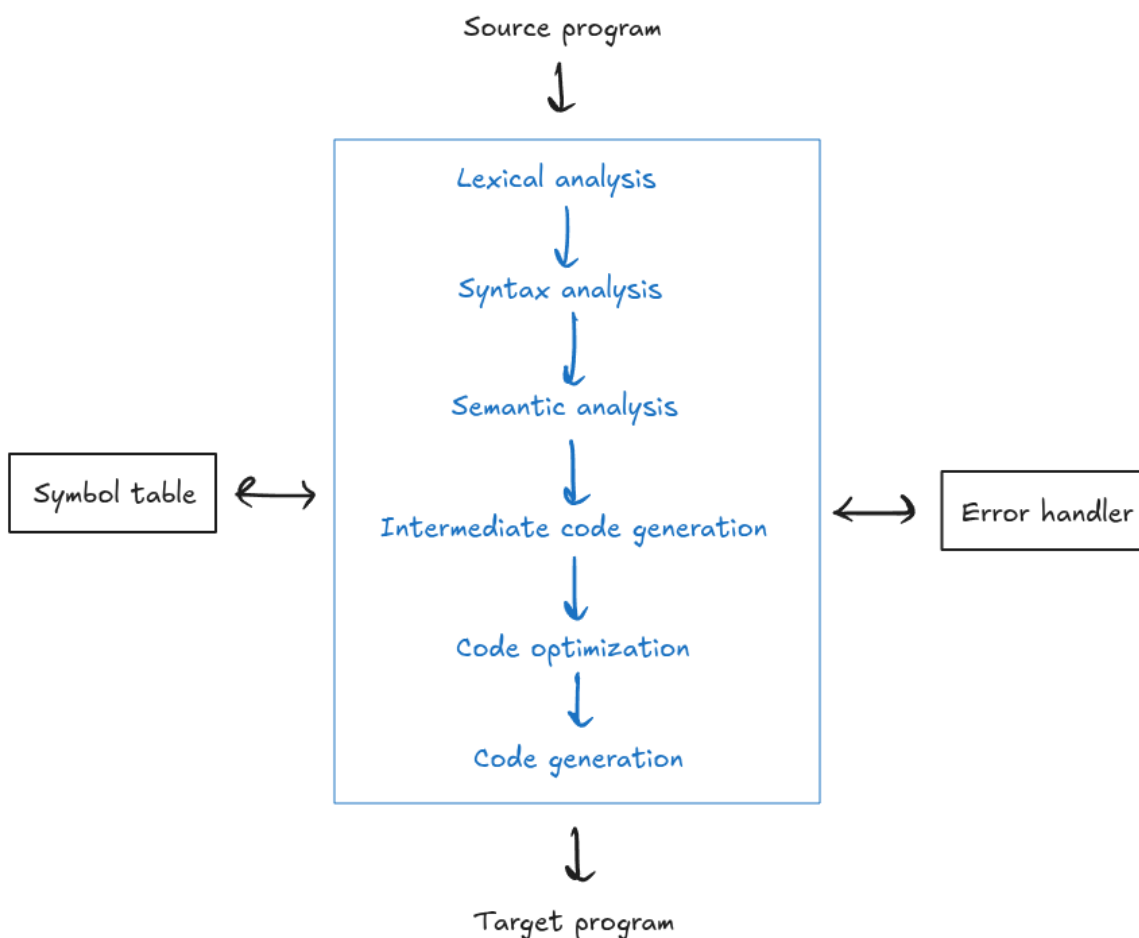


Figure 2: Stages of Compilation

1	position	...
2	initial	...
3	rate	...

SYMBOL TABLE

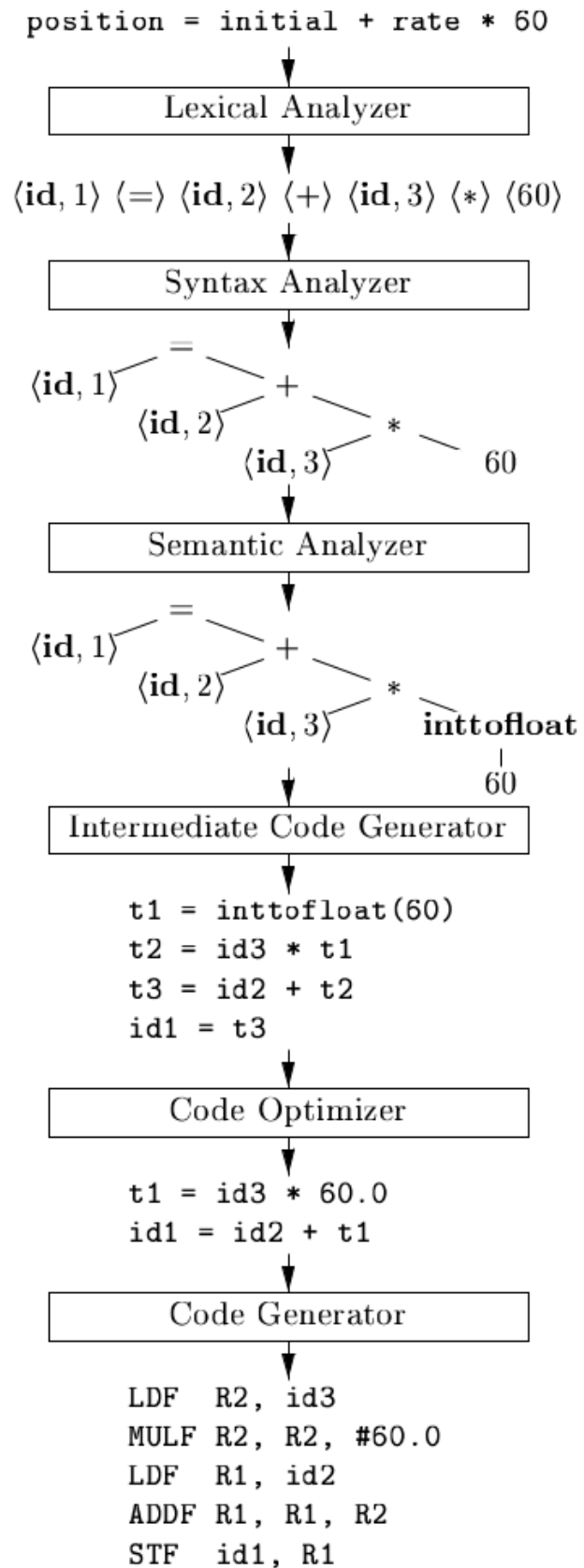


Figure 3: Stages of compilation - Example

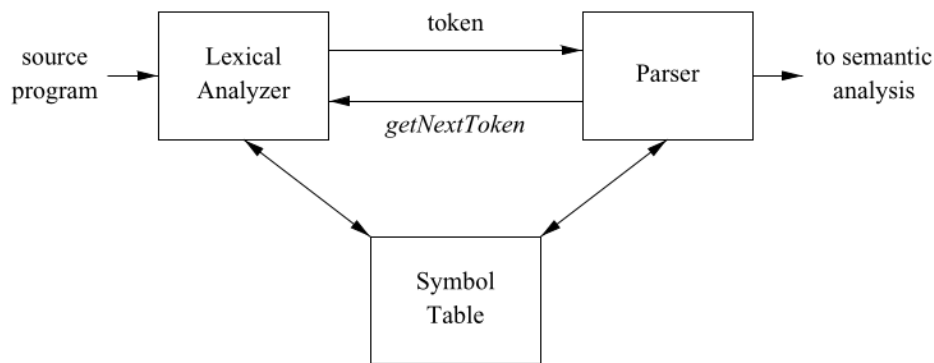


Figure 4: Lexical analyzer

letter = A-Z | a-z

digit = 0-9

identifier = letter . (letter + digit)*

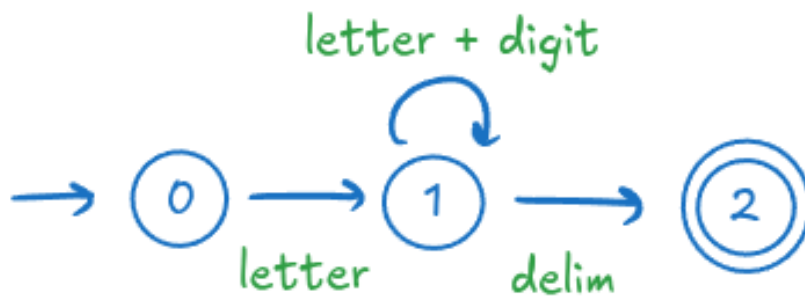


Figure 5: Recognizers for identifiers

- Delimiter is any value which can be used to end a token.
- It may be a space, assignment operator, semicolon, newline character or anything else according to the definition of the language.

Pseudo code for recognizing identifiers

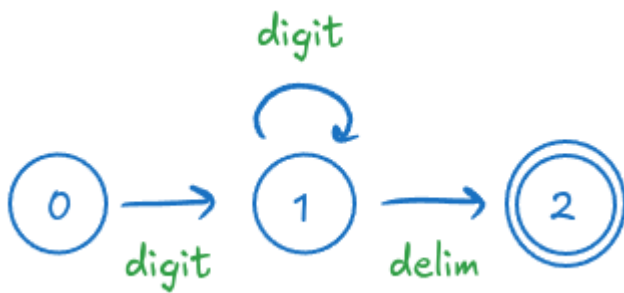


Figure 6: Recognizer for numbers

state 0:

```
c = getchar()
if c is letter
    goto state 1
else fail
```

state 1:

```
c = getchar()
if c is letter or c is digit
    goto state 1
else if c is delimiter
    goto state 2
else fail
```

state 2:

```
accept identifier
```